From Turing to Transformers

A Timeline of Artificial Intelligence, 1950 to Today

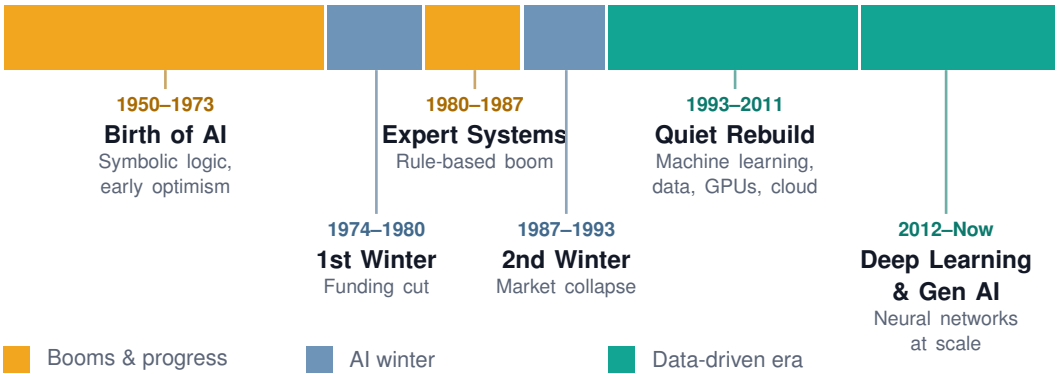
Key milestones, the AI winters, and the deep learning and cloud computing revolution



Sharon Ochieng | AIML-500 | 10/29/2026

THE BIG PICTURE

Seven Decades of Booms and Winters



The Foundations of AI

1943

First artificial neuron

Warren McCulloch and Walter Pitts model a neuron mathematically, the seed of neural networks.

1950

The Turing Test

Alan Turing asks “Can machines think?” and proposes the imitation game in “Computing Machinery and Intelligence.”

1956

Dartmouth Workshop

John McCarthy, Marvin Minsky, Claude Shannon and Nathaniel Rochester coin “artificial intelligence.” AI becomes a field.

1958

The Perceptron

Frank Rosenblatt builds a machine that learns to recognize simple patterns, an early neural network.

1966

ELIZA chatbot

Joseph Weizenbaum at MIT creates the first chatbot, which imitates a therapist using simple pattern matching.

Era of Symbolic AI: intelligence as logic and rules written by humans (Bhat, 2025).

The First AI Winter



1974–1980

Hype outran the hardware and the math

Causes

- Early promises (machine translation, general problem solvers) were not delivered
- Minsky & Papert's *Perceptrons* (1969) exposed limits of simple neural nets
- UK Lighthill Report (1973) judged AI a failure
- Computers lacked the memory and speed AI needed

Impacts

- DARPA and UK governments cut AI research funding
- Neural network research nearly stopped for a decade
- Researchers rebranded work to avoid the "AI" label

Expert Systems Bring AI to Business

1980

XCON at DEC

An expert system with thousands of if-then rules configures computer orders, saving Digital Equipment Corp. millions per year.

1982

Japan's Fifth Generation

Japan launches a major national project to build intelligent computers, sparking new funding in the U.S. and U.K.

1986

Backpropagation

Rumelhart, Hinton and Williams popularize the method that lets multi-layer neural networks learn, still used in deep learning today.

Rule-based and expert systems defined the 1980s (Bhat, 2025).

The Second AI Winter



1987–1993

Brittle rules could not scale to the real world

Causes

- Specialized Lisp machine market collapsed in 1987 as cheaper desktop PCs caught up
- Expert systems were costly to maintain and broke on unexpected cases
- Japan’s Fifth Generation project ended in 1992 short of its goals

Impacts

- Hundreds of AI companies closed or were sold
- Funding agencies grew skeptical of AI claims
- Focus shifted to practical, statistical machine learning

Data, GPUs, and the Cloud Set the Stage

1997

Deep Blue

IBM's chess computer defeats world champion Garry Kasparov using brute-force search.

1999

The GPU

Nvidia releases the GeForce 256, marketed as the first GPU. Its parallel chips later power deep learning.

2006

Cloud + deep nets

Amazon Web Services launches rentable cloud computing. Geoffrey Hinton revives "deep" neural networks.

2009

ImageNet

Fei-Fei Li releases a dataset of millions of labeled images, giving AI the data it needed to learn vision.

2011

Watson & Siri

IBM Watson wins *Jeopardy!* and Apple launches Siri, bringing AI to the public.

Cloud computing made massive compute affordable to rent instead of buy.

The Deep Learning Revolution

2012

AlexNet

Krizhevsky, Sutskever and Hinton win ImageNet by a huge margin with a neural net trained on two Nvidia GPUs.

2014

GANs & DeepMind

Ian Goodfellow invents GANs, which generate images. Google acquires DeepMind.

2016

AlphaGo

DeepMind's AlphaGo beats Go champion Lee Sedol 4–1, a feat experts expected to be a decade away.

2017

The Transformer

Google researchers publish "Attention Is All You Need," the architecture behind every major chatbot today.

The "AlexNet moment": hardware finally caught up with neural network theory (Putchuon, 2024).

The Generative AI Era

2018

GPT & BERT

OpenAI's GPT-1 and Google's BERT show transformers can learn language from huge text datasets.

2020

GPT-3 & AlphaFold

GPT-3 writes fluent text. AlphaFold 2 predicts protein structures, solving a 50-year biology problem.

2022

ChatGPT

OpenAI releases ChatGPT on Nov. 30, and generative AI goes mainstream almost overnight.

2023

The AI race

GPT-4 (OpenAI/Microsoft), Claude (Anthropic) and Gemini (Google) compete. Nvidia becomes the key chip supplier.

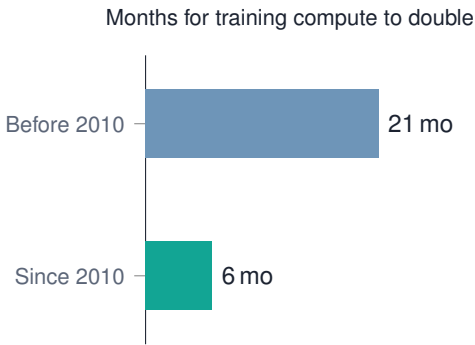
2024+

Nobels & reasoning

Hinton, Hassabis and Jumper win Nobel Prizes. "Reasoning" models and AI agents emerge.

Generative AI creates new content rather than only predicting from existing data (Bhat, 2025).

Computing Power Is Doubling Faster Than Ever



3.5×

faster growth in the compute used to train notable AI systems since deep learning took off

Benchmarks have shifted the same way, from beating humans at chess (1997) and Go (2016) to writing, coding and passing professional exams.

Data: Giattino & Samborska (2025), Our World in Data; Roser (2022).

MAJOR PLAYERS

The People and Companies Behind AI

Pioneers

1. **Alan Turing**
   Defined machine intelligence (1950)
2. **John McCarthy**
   Coined “artificial intelligence” (1956)
3. **Hinton, LeCun & Bengio**
   Deep learning pioneers, 2018 Turing Award
4. **Fei-Fei Li**
   Created ImageNet for computer vision
5. **Demis Hassabis**
   DeepMind founder: AlphaGo, AlphaFold

Leading companies

1. **OpenAI**
   GPT models and ChatGPT
2. **Google / DeepMind**
   Transformer, Gemini, AlphaFold
3. **Nvidia**
   GPUs that train nearly all modern AI
4. **Microsoft & Amazon**
   Cloud platforms (Azure, AWS) that host AI
5. **Anthropic & Meta**
   Claude; open Llama models

WHERE WE ARE TODAY

Narrow AI Today, AGI Still Ahead

Narrow AI (exists now)

Excels at specific tasks: recommending products, recognizing faces, translating, generating text. Every system on this timeline is narrow AI.

AGI (not yet achieved)

Artificial General Intelligence would match humans across any intellectual task. Experts strongly disagree on when, or whether, it will arrive.

Lessons from the timeline

- Progress needed theory, data and hardware together; the 2012 breakthrough came when all three aligned.
- Winters followed overpromising, a caution for today's hype.
- Cloud computing and GPUs turned AI from a lab experiment into an industry.

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AI tool disclosure: Claude (Anthropic) was used to help research milestones and draft this timeline. All facts were reviewed against the sources above.